
CSC 412 – Operating Systems

Lab Session 03, Fall 2025

Tuesday, September 23rd and Thursday, September 25th, 2025

What this Lab Session Is About

This lab is about file I/O in C.

1 Task 1: Look (Again?) at the Handout Code

The handout code was already posted on BrightSpace on the “Code Samples” section, and it is joined to this lab as a quick reference. It goes over the basic forms of file I/O in C. Make sure that you understand what is going on before proceeding to the next section.

Please note that some sections of the program will not run the first time around, because some folders are missing, or some data haven’t got the right format. Check in the debugger where the program crashes, read the comments, and fix the problem.

2 Task 2: Read and Write Lists of Words

The data to read come in a text file in which words are listed by categories. The first line of the text file gives the number of categories. The data for each category are then listed on separate lines, starting with the name of the category (e.g. `articles`, `verbs1`, `nouns4`; this name doesn’t contain any blank character), then the number of words in the category, and finally the words of the category (again, these words cannot contain spaces). An example of a data file would be

```
4
adjectives1 6 happy sad moody grumpy cheerful joyful
verbs1 4 eat chew digest dine
verbs2 5 run walk stroll jump hop
nouns1 4 street road alley avenue
```

2.1 Task 2.1 Flatten the list

Read one such data file and write back to a file a flat list of all the words in the input file.

2.2 Task 2.2 Merge two lists

Your task is to read two such data files and write to a file the result of merging the data of these two lists. For example, if the first file contains the data listed earlier and the second data file contains

3

adjectives2 4 tall short long wide

verbs1 3 devour digest swallow

nouns1 2 boulevard lane

then the file resulting from the merging would store

adjectives1 6 happy sad moody grumpy cheerful joyful

verbs1 6 eat chew digest dine devour swallow

verbs2 5 run walk stroll jump hop

nouns1 6 street road alley avenue boulevard lane

adjectives2 4 tall short long wide